

ECE 3210 hw02

1. Consider the system $y(t) = x(2t)$ with input $x(t) = |t|$. Let $T = 1$ and define

$$\tilde{y}(t) = H\{x(t - T)\}.$$

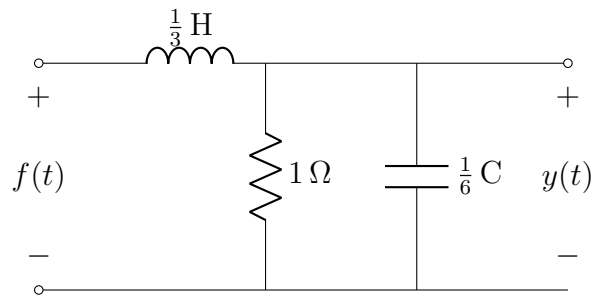
- (a) Find $y(t)$, $\tilde{y}(t)$, and $y(t - T)$. Plot each of the three signals on the same axes.
- (b) Is the system time invariant? Hint: compare $\tilde{y}(t)$ and $y(t - T)$. If they are not equal, then the system is not time invariant.

2. Consider the real LTIC system

$$(D^2 + 5D + 6)y(t) = (D^2 + 7D + 11)x(t).$$

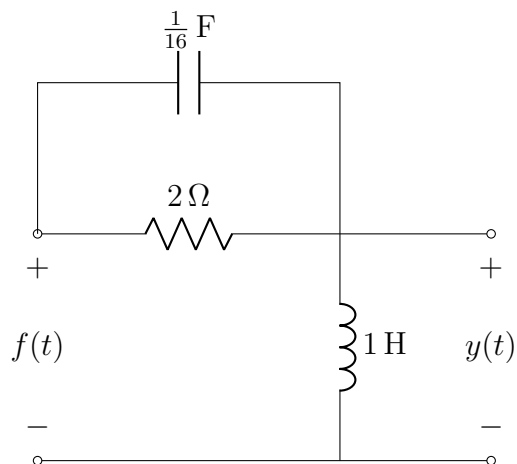
- (a) What is the order n of the system?
- (b) Find the characteristic equation $Q(\lambda) = 0$.
- (c) Find the quasi-zero-input response $y_n(t)$.
- (d) Find the b_n term.
- (e) Find the impulse response $h(t)$.

3. For the circuit below, with input $f(t)$ and output $y(t)$, answer the following questions.



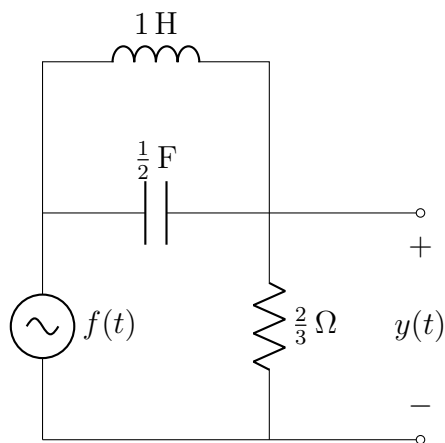
- (a) Determine the ODE governing the system.
- (b) Determine the zero-input response $y_n(t)$.
- (c) Determine the impulse response $h(t)$.

4. For the circuit below, with input $f(t)$ and output $y(t)$, answer the following questions.



- (a) Determine the ODE governing the system.
- (b) Determine the zero-input response $y_n(t)$ (as if we were trying to find $h(t)$).
- (c) Determine the impulse response $h(t)$.

5. For the circuit below, derive the input-output relation and answer the following questions.



- (a) Determine the ODE governing the input-output relation of the circuit.
- (b) Given $y(0^-) = 0$ and $y'(0^-) = 1$, determine the zero-input response $y_n(t)$.
- (c) Determine the impulse response $h(t)$.

Answer Key

1(a) $y(t) = 2|t|$, $\tilde{y}(t) = |2t - 1|$, $y(t - T) = 2|t - 1|$

1(b) Not time-invariant

2(a) $n = 2$

2(b) $Q(\lambda) = \lambda^2 + 5\lambda + 6 = 0$

2(c) $y_n(t) = e^{-2t} - e^{-3t}$

2(d) $b_n = 1$

2(e) $h(t) = \delta(t) + (e^{-2t} + e^{-3t})u(t)$

3(a) $y''(t) + 6y'(t) + 18y(t) = 18f(t)$

3(b) $y_n(t) = \frac{1}{3}e^{-3t} \sin(3t)$

3(c) $h(t) = 6e^{-3t} \sin(3t)u(t)$

4(a) $y''(t) + 8y'(t) + 16y(t) = f''(t) + 8f'(t)$

4(b) $y_n(t) = te^{-4t}$

4(c) $h(t) = \delta(t) - 16te^{-4t}u(t)$

5(a) $y'' + 3y' + 2y = f'' + 2f$

5(b) $y_n(t) = e^{-t} - e^{-2t}$

5(c) $h(t) = \delta(t) + (3e^{-t} - 6e^{-2t})u(t)$